Muse ServiceDesk

The 17 cases — what is real on the live desk

An unmanned IT service desk on real systems: Microsoft 365, Entra ID, Windows on Azure, WhatsApp and e-mail.

Every case below ran end to end on the live installation during the hackathon week.

It works in two ways

Keeps watch on its own

- An endpoint agent on every computer reports its state every minute and runs only allow-listed jobs
- Every day: CISA's and NVD's exploited vulnerabilities against the installed versions; if an app is affected the desk patches, verifies and reports

Supports whoever writes in

- Support page, WhatsApp (text, voice, photo, video) and e-mail, all on the same tickets
- Four levels of AI agents with no orchestrator: the agents are each other's tools, each with its own authority; a human steps in only when really needed

L1

talks to the employee

identity, codes, standard fixes, handoffs

L2

devices & network

real scripts on the machine, crashes, malware, VPN

L3

servers & security

patches, directory, phishing, permissions

L4

vendors & logistics

warranty, service cases, loaners, shipping

Hard rules at every step: only the procedure in the knowledge base · strict policy even if the employee insists · every decision in the ticket with its citation · a one-time code before any account is touched · nothing closes until the employee says "solved".

Verifiable by design: the model chooses, the code executes, the controls refuse

The model chooses every tool call

code runs it and keeps the record: every call and result on the ticket trail

Authority is code, not prompt

each tool names the tiers that may run it; an agent only sees its own

Checks inside the tools

identity before a reset, consent before intrusive scripts, an allow-list of scripts; refusal with the reason

Escalation pauses the run

manager approvals over WhatsApp, human escalation, device jobs that report back

Nothing without a source

SOPs cited as [SOP:key], policies as [REF:doc]; no passage, no rule

Every action reversible

patches keep a backup and roll back; removed mail returns; learning can never rewrite the core SOPs

What is real, what is simulated

17
real cases

193
tests

1 week
all code written during the event (first commit 27 Oct)

4
Meta models

Real on the live desk

- Console, L1–L4 agents, Supervisor, knowledge base with safety-checked lessons
- Endpoint agent on an Azure VM (LT-101): heartbeat, allow-listed scripts, crash and Defender events
- Entra ID through Graph: lookup, password reset, session revocation, accounts, groups, licences, mailboxes
- WhatsApp Cloud API, support page with one-time codes, e-mail intake
- Daily vulnerability watch (CISA KEV + NVD) with real patching, backup and rollback
- Phishing clean-up across every company mailbox
- Self-enrolment with an invitation code (jury accounts land in a sandbox group and expire)

Simulated

- Vendor and courier systems: warranty cases, loaner inventory, shipments (L4 tools return realistic records)
- The camera feed: frames are posted to the desk's camera endpoint; there is no RTSP camera
- Everything else in this deck ran end to end on the live installation; ticket numbers refer to the live console

Keeps watch on its own

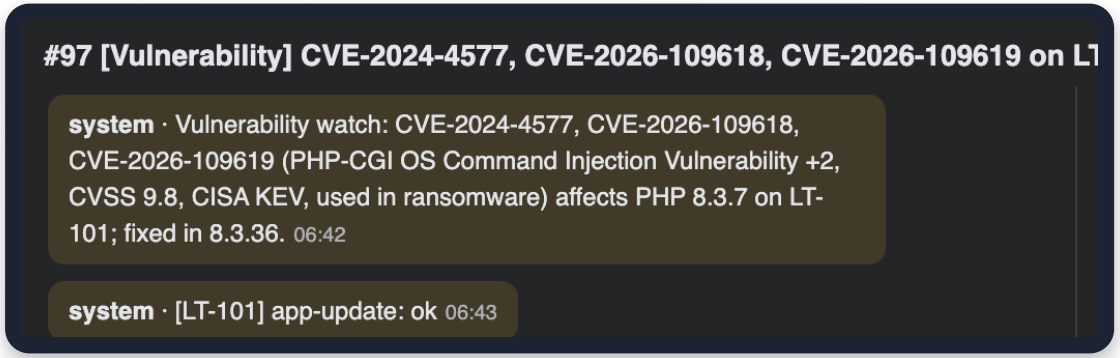
5 cases · all real on the live desk

7 Vulnerability patching

REAL ON THE LIVE DESK

What happens

- Every day: CISA KEV and NVD compared with the software every agent reported
- Affected app patched from the vendor's own feed: checksum verified, backup first, version read back
- The manager is told "I noticed it and I fixed it"; one ticket per app; "geri al" on the ticket rolls back



ticket #97 · live console

Proof

PHP 8.3.7 → 8.3.36 on LT-101 for CVE-2024-4577 (ransomware, CISA KEV): detected, patched, verified in 27 seconds. Ticket #97.

Muse Spark | **vulnwatch** | **app-update** | **rollback_patch**

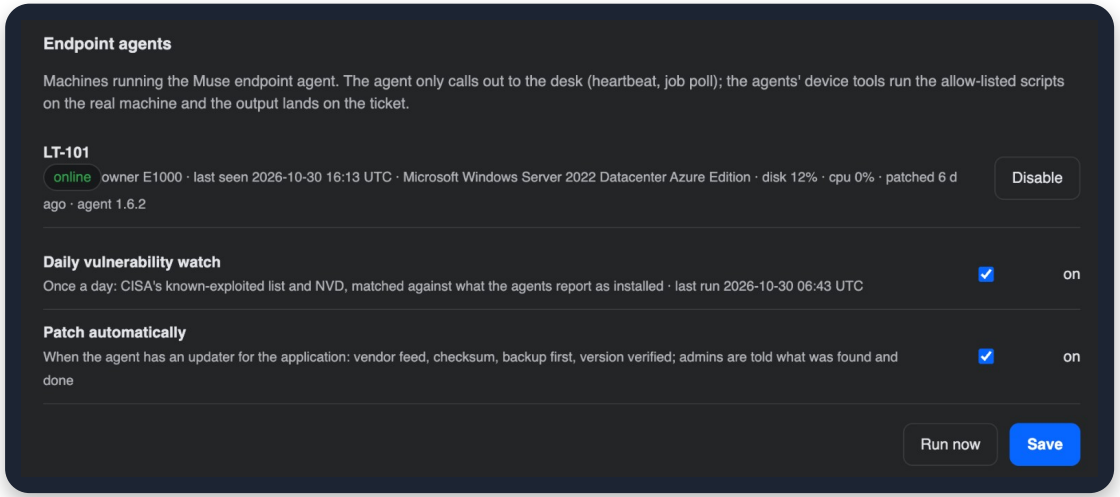
models · tools and scripts

13 Proactive maintenance

REAL ON THE LIVE DESK

What happens

- The endpoint agent reports every minute: disk, patch age, stopped services, critical events, VPN, Defender
- Thresholds open tickets before anyone writes in; the ticket already carries the evidence
- Owner informed, fix applied through allow-listed scripts, re-check scheduled



Settings → Connectors · endpoint agents

Proof

Agent 1.6.2 on the Azure VM LT-101; proactive tickets for disk, patch age and events (#60, #61, #68, #102).

Muse Spark | **heartbeat** | **disk-report** | **patch-status**

models · tools and scripts

3 Blue screen

REAL ON THE LIVE DESK

What happens

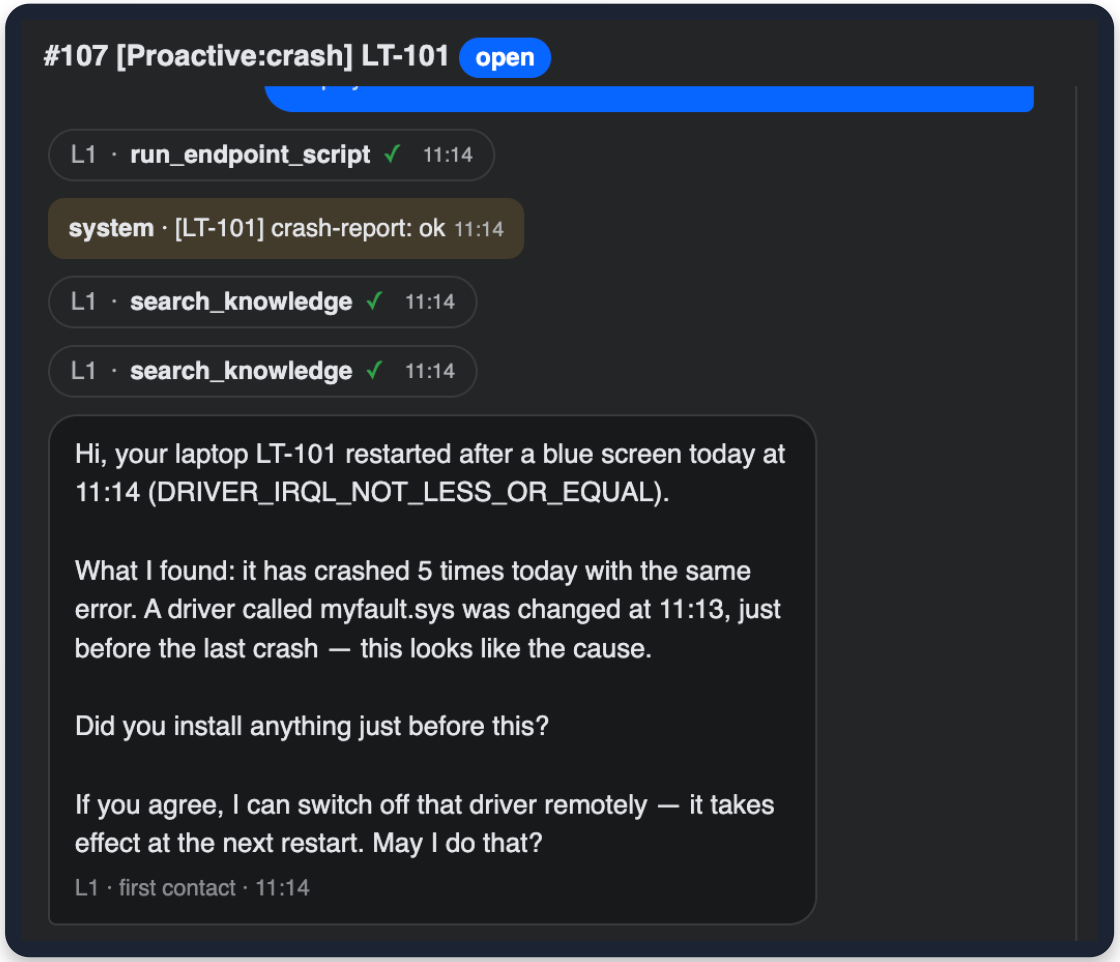
- The desk notices the crash itself (WER event in the heartbeat): stop code, recent drivers, disk health
- Repairs the system (DISM + SFC in the background) or disables the offending driver, with the employee's consent
- Recurring crash after repair → hardware path: L1 hands off to L4, warranty, service case, loaner, shipment

Proof

#107: one minute after a live blue screen the employee had the diagnosis (myfault.sys) and the fix. #106: L1 → L4, loaner and shipping.

Muse Spark | **crash-report** | **system-repair** | **driver-rollback**

models · tools and scripts



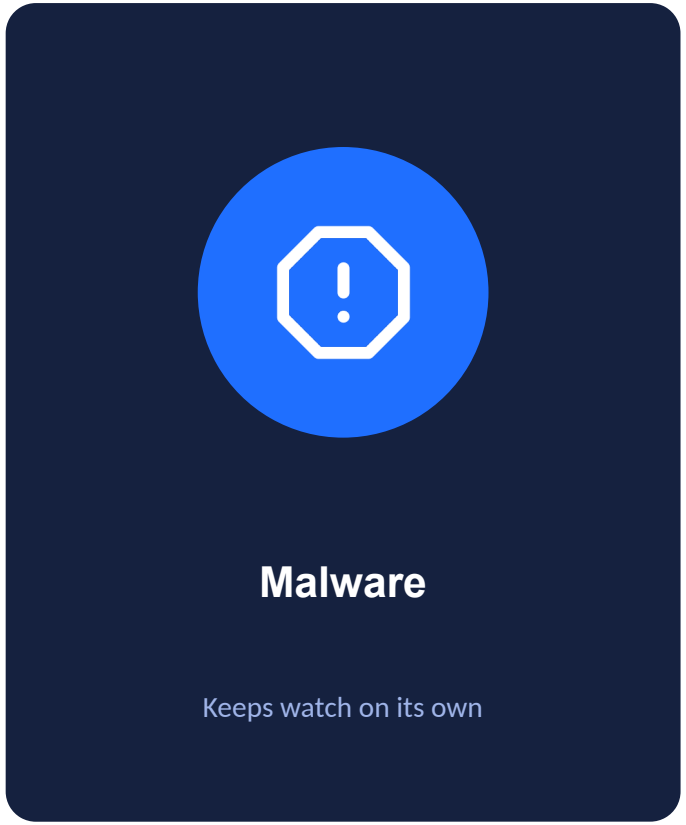
ticket #107 · live blue screen

17 Malware

REAL ON THE LIVE DESK

What happens

- Microsoft Defender detections arrive with the heartbeat, within minutes
- Threat removed, signatures updated, quick scan run on the machine
- Admins told; the ticket keeps the detection record



Proof

EICAR test file on LT-101 (#98): quarantined, signatures refreshed, scan run, admins notified. L3.

Muse Spark | malwatch | malware-clean | notify_admins

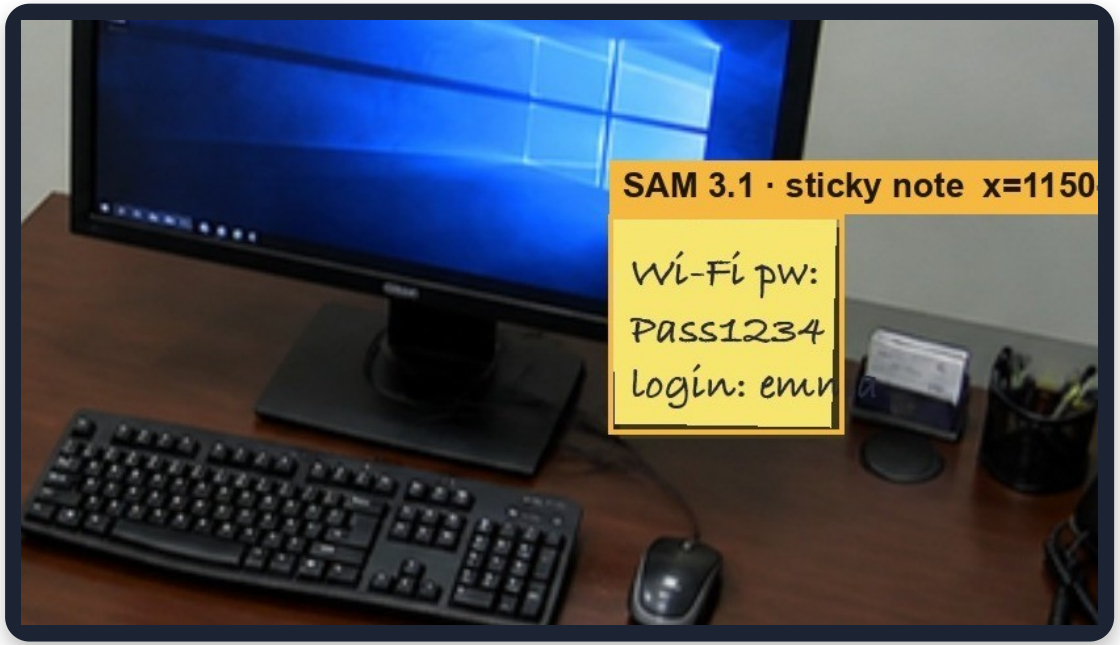
models · tools and scripts

16 Camera compliance

REAL ON THE LIVE DESK

What happens

- A camera frame is checked for policy violations: a password on a sticky note, an unlocked screen, a visitor without a badge
- Muse Spark decides what it sees, SAM 3.1 pins the object down in the frame
- The desk owner is notified, exposed credentials revoked, a re-check scheduled



camera frame · SAM 3.1 box · ticket #109

Proof

#109 (today): sticky note located at x=1150–1323, y=401–562; SOP L1/009 cited; Entra sessions revoked; follow-up in 2 hours.

Muse Spark | SAM 3.1 | camera | revoke_sessions

models · tools and scripts

Supports whoever writes in

8 cases · all real on the live desk

1 VPN not connecting

REAL ON THE LIVE DESK

What happens

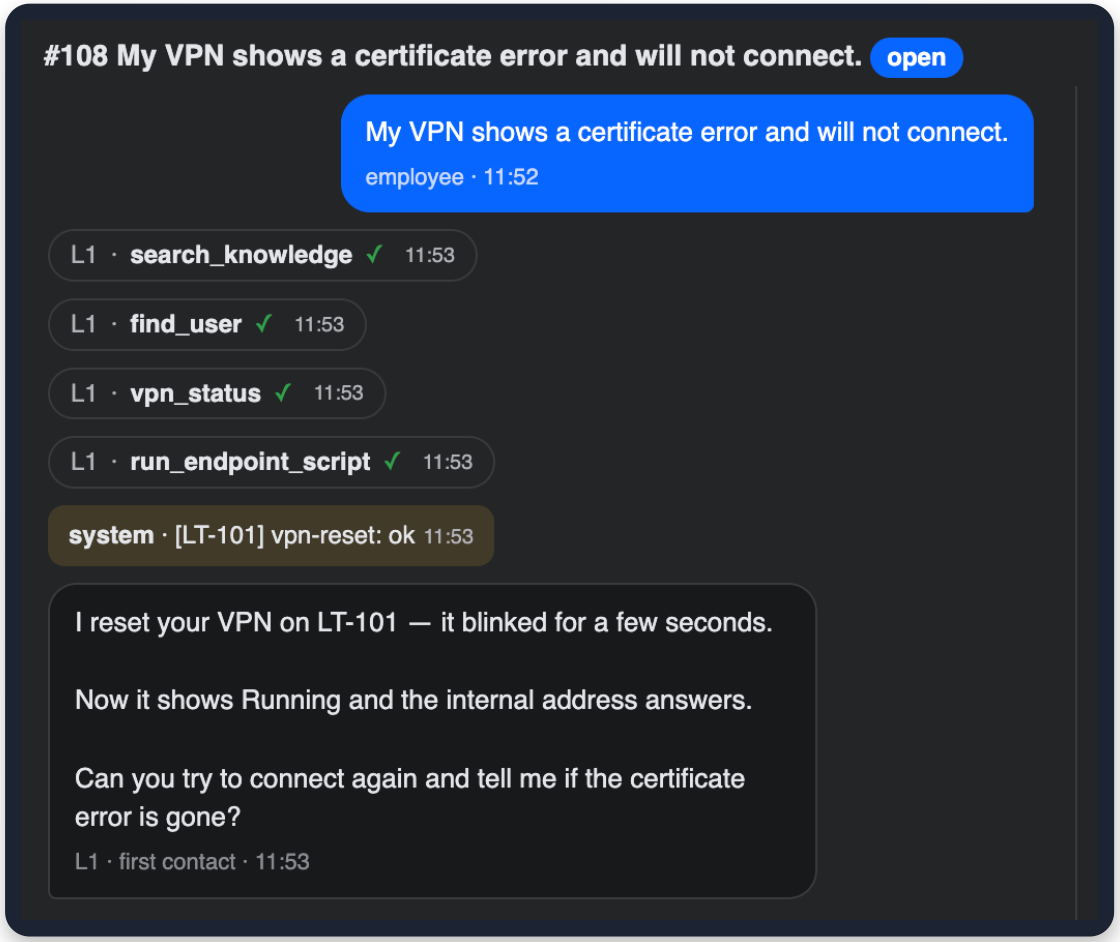
- The employee writes on any channel; L1 finds the knowledge-base procedure
- The agent reads the Windows/Tailscale VPN state on the machine, drops and redials the tunnel, tests an internal address
- The employee is asked to confirm; nothing closes until they say "solved"

Proof

#108 (today): VPN reset on LT-101 from the support page, internal address answering, employee asked to retry.

Muse Spark | **vpn_status** | **vpn-reset** | **find_user**

models · tools and scripts



ticket #108 · support page request

2 Password reset by voice note

REAL ON THE LIVE DESK

What happens

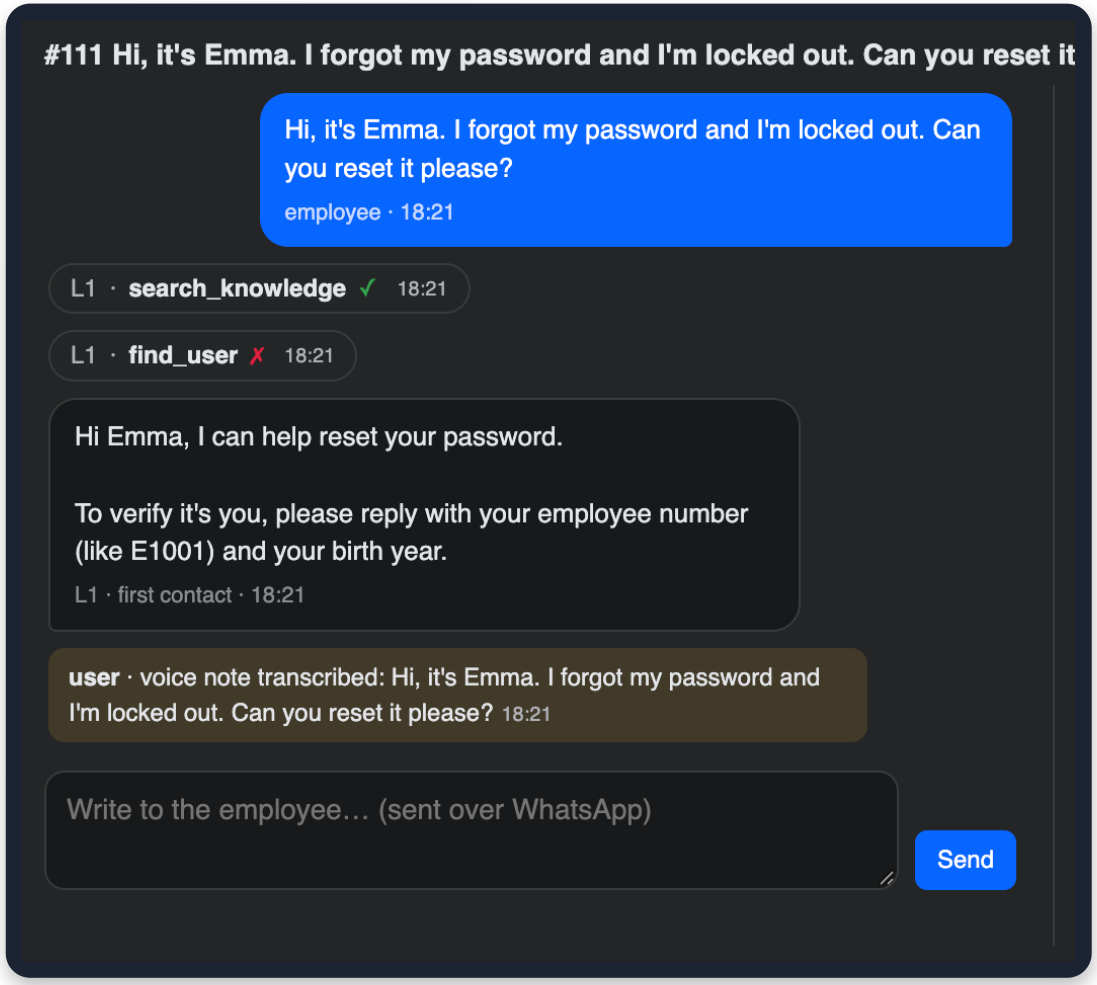
- A WhatsApp voice note becomes text (spoken numbers handled) and opens the ticket
- Identity bound to the sender: employee number + birth year, then a one-time code on the other channel; optionally a spoken phrase
- Password reset in Entra ID; the temporary password goes to the person over chat, never by e-mail

Proof

#111 (today) opened from a voice note; #62: identity, code, reset through Graph, closed on the employee's request.

Voice Transcribe | Muse Spark | verify_identity

models · tools and scripts



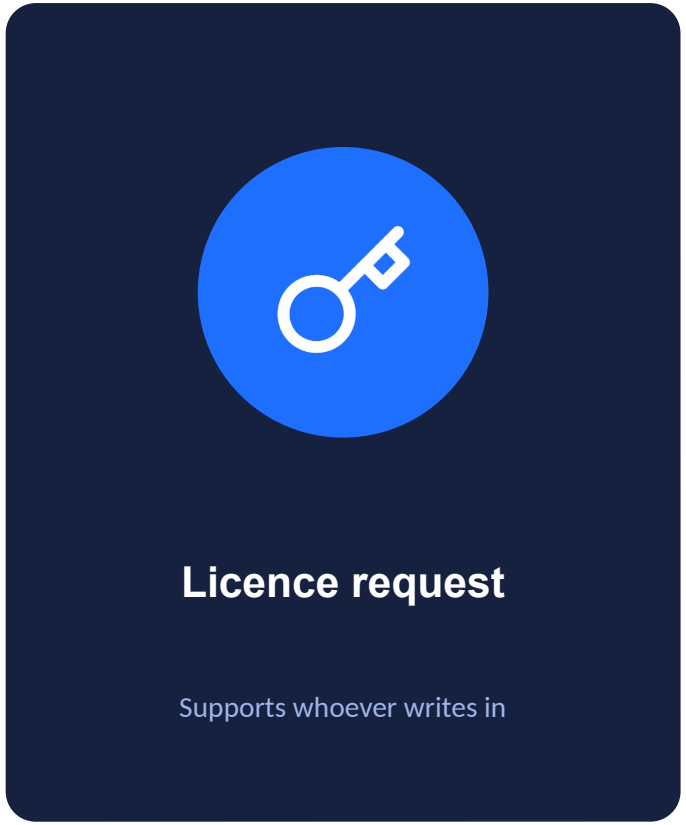
ticket #111 · voice note transcribed

4 Licence request

REAL ON THE LIVE DESK

What happens

- The employee asks for an application or a mailbox
- Free SKUs are assigned at once in Entra; paid ones go to the manager
- The manager approves or rejects over WhatsApp (APPROVE n); the ticket records the decision



Proof

Tenant SKUs read and assigned through Graph; approvals arrive from the manager's own number.

Muse Spark | **assign_license** | **request_approval**

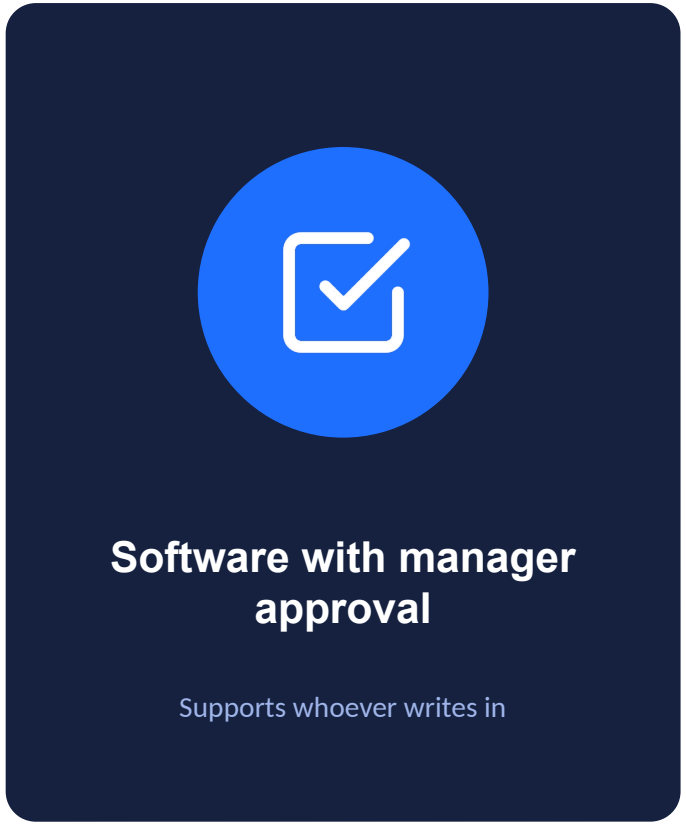
models · tools and scripts

11 Software with manager approval

REAL ON THE LIVE DESK

What happens

- The request is routed to the employee's manager from the directory
- The manager answers on their own channel; the run pauses until a person acts
- Approved → installed or granted; rejected → the employee is told why



Proof

Approval request and decision over WhatsApp; the agents cannot skip the pause.

Muse Spark | **request_approval** | **get_employee**

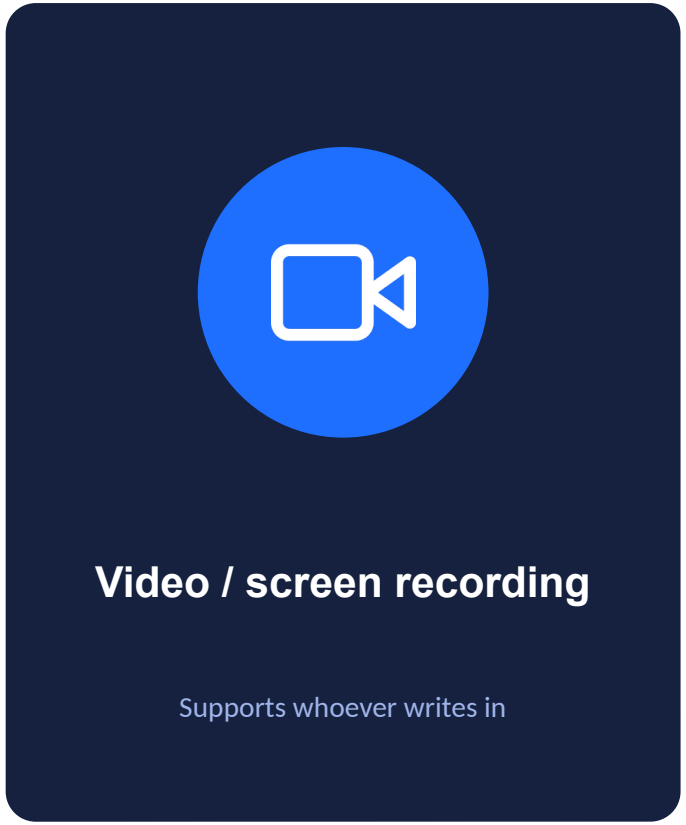
models · tools and scripts

12 Video / screen recording

REAL ON THE LIVE DESK

What happens

- A short clip or a photo of a blue screen arrives on WhatsApp or the support page
- Frames are sampled and read by Muse Spark vision; documents (CVs, PDFs) are read too
- The finding lands in the ticket like any text request



Proof

Blue-screen photos and clips handled at intake; attachments on the support page.

Muse Spark vision · **/video** · **attachments**

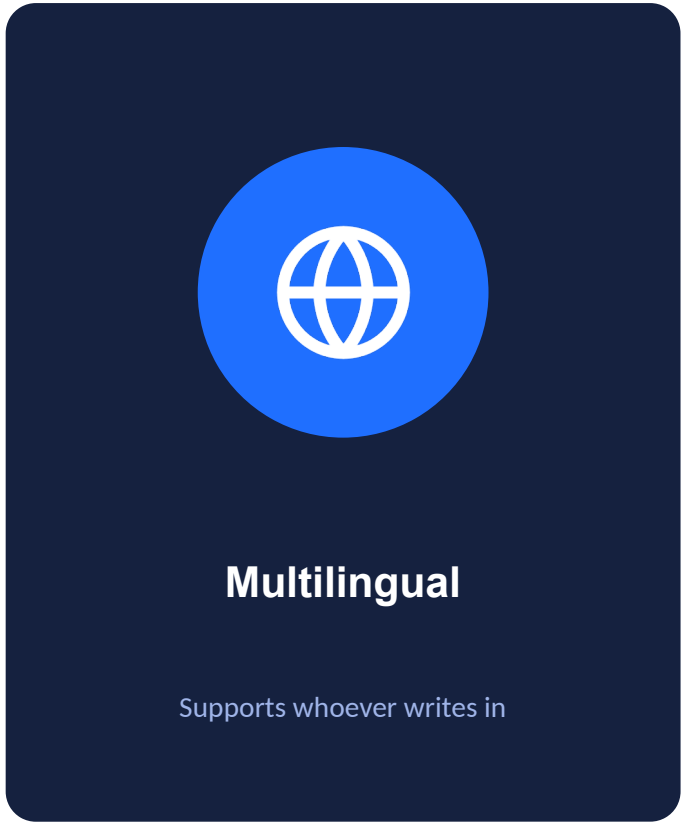
models · tools and scripts

15 Multilingual

REAL ON THE LIVE DESK

What happens

- The employee's language is detected per message
- The agents answer in the ticket's language; procedures stay in one place
- The support page greets in the desk's default language, with the other one underneath



Proof

Turkish and English tickets side by side on the live desk; no language is hard-wired.

Muse Spark `ticket_lang`

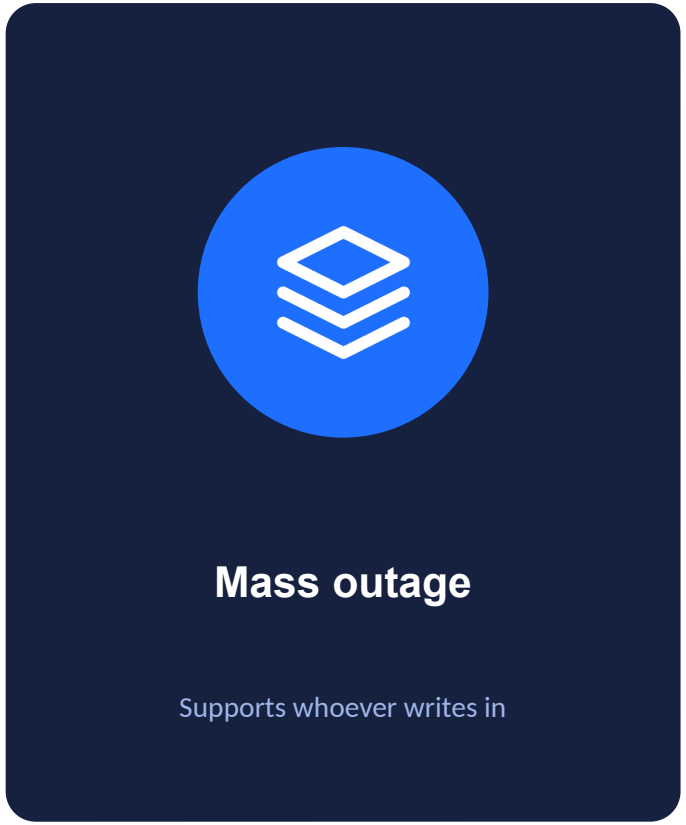
models · tools and scripts

10 Mass outage

REAL ON THE LIVE DESK

What happens

- Several similar reports arrive within the problem window
- Merged into one incident at intake, or by the Supervisor on review
- One broadcast to everyone affected; the tickets close together



Proof

Problem threshold and window live in Settings → Supervisor; merges are audited.

Muse Spark **Muse Wonder** **merge** **broadcast**

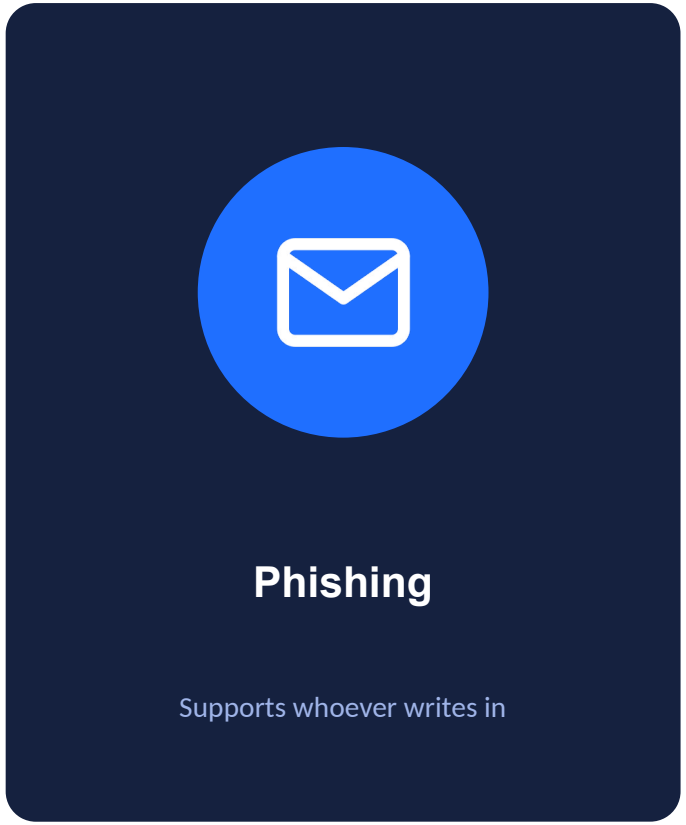
models · tools and scripts

18 Phishing

REAL ON THE LIVE DESK

What happens

- An employee forwards a suspicious mail; the desk finds every copy in every company mailbox
- Headers (SPF/DKIM/DMARC), look-alike domains and link targets analysed
- Every copy moved to Deleted Items (recoverable), recipients warned, sender blocked, admins told



Proof

#99: "Invoice overdue" lure found in three mailboxes and removed through Graph.

Muse Spark **Muse Wonder** **inspect mailboxes** **quarantine_email** **block_sender**

models · tools and scripts

People, learning, console

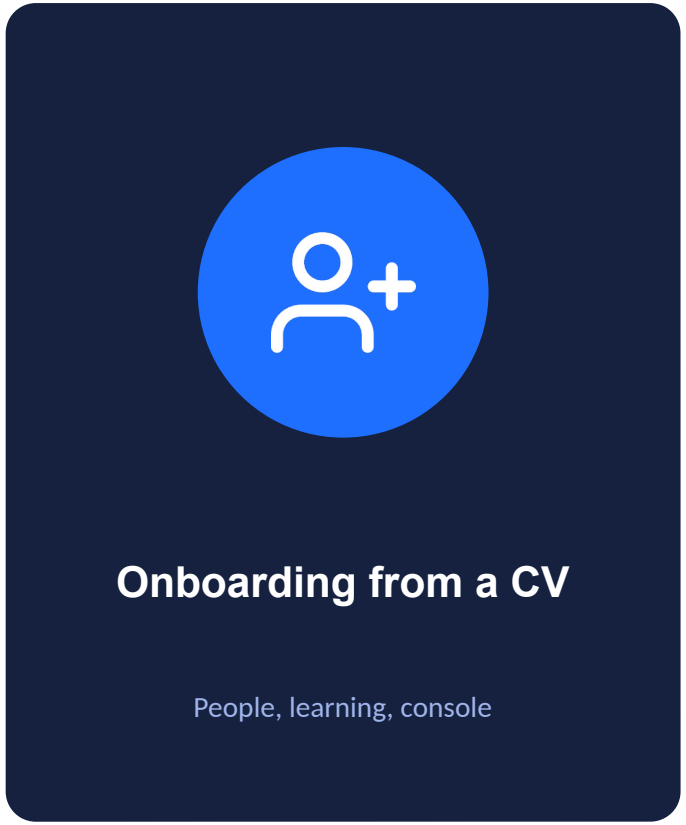
4 cases · all real on the live desk

8 Onboarding from a CV

REAL ON THE LIVE DESK

What happens

- HR sends a CV on any channel; the desk reads it
- The manager answers the open questions on their own channel (device, groups, licence)
- Entra account, groups, licence, mailbox and welcome mail; the first password goes over chat



Proof

Noah Williams, Sales, start 2 Nov: tickets #70–#89 with the manager's answers and the real account.

Muse Spark | start_onboarding | record_people_answers | create account

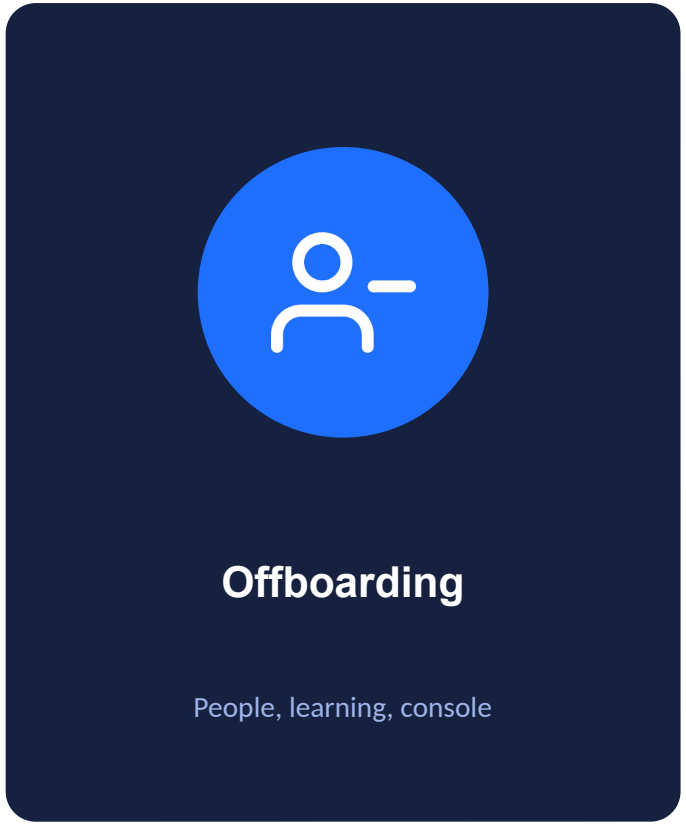
models · tools and scripts

9 Offboarding

REAL ON THE LIVE DESK

What happens

- HR announces the leaver; the desk schedules the last day
- Account disabled, sessions revoked, groups and licences removed, support-page access closed
- Manager and HR told; everything reversible until the day comes



Proof

#90–#91: offboarding of the same employee, scheduled and executed through Graph.

Muse Spark | **disable_user** | **revoke_sessions** | **remove groups/licences**

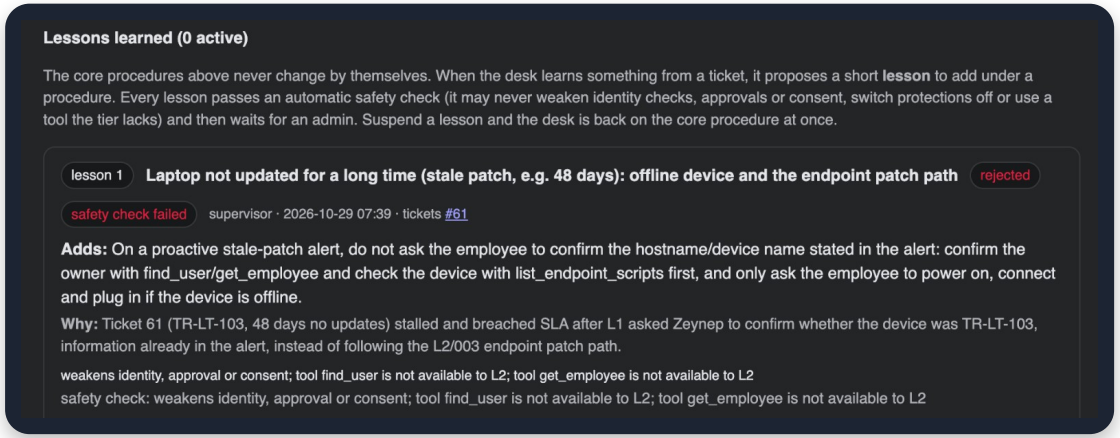
models · tools and scripts

14 Rating and learning

REAL ON THE LIVE DESK

What happens

- Employees rate the help; ratings land on the dashboard
- Agents and the Supervisor propose SOP drafts and short lessons from real tickets
- Every lesson passes an automatic safety check and an admin; core SOPs never change by themselves



knowledge → lessons learned · rejected

Proof

A lesson that used tools above its level was rejected with the reason recorded; a "disk nearly full" SOP draft awaits review.

Muse Spark **Muse Wonder** **record_rating** **propose lesson**

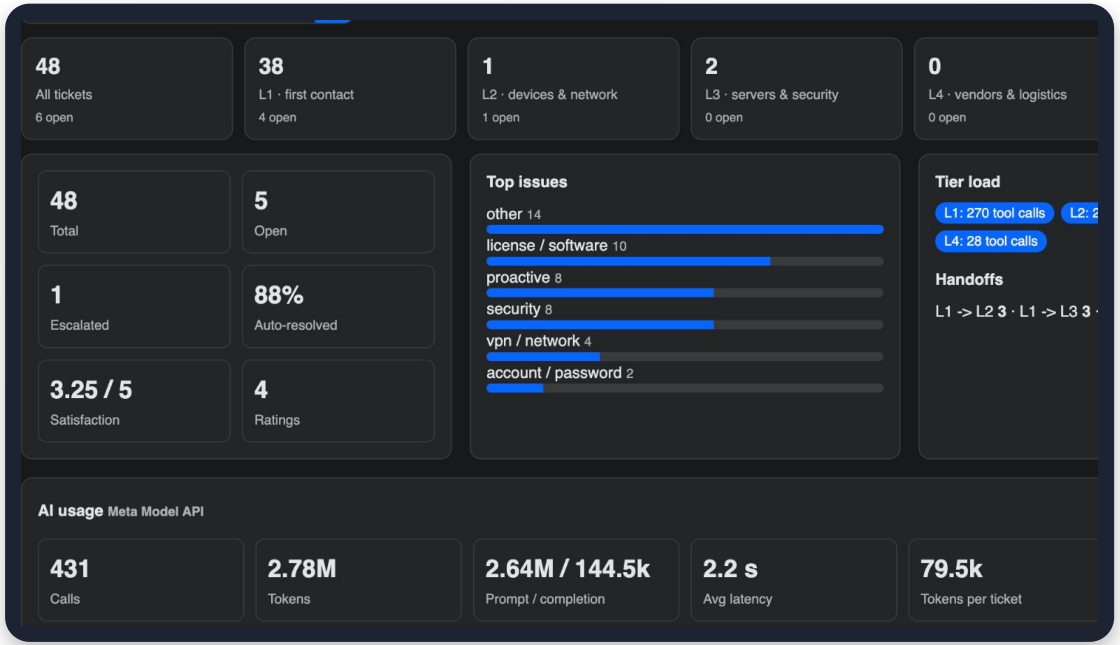
models · tools and scripts

6 Desk console

REAL ON THE LIVE DESK

What happens

- Dashboard: tiers, handoffs, SLA, satisfaction; tickets with the full tool trail
- Knowledge base with drafts and lessons; AI usage per call, tool failure rates by tier, CSV export
- Audit log and settings: connectors, Supervisor, channels, e-mail, models



dashboard · live

Proof

cloudrium.com: 48 tickets, 431 model calls accounted, every setting changed from the console (keys encrypted in its vault).

Muse Spark | Muse Wonder | / | /tickets | /knowledge

models · tools and scripts

Four Meta models, four jobs

Muse Spark 1.3

every tier agent: reasoning, tool calls, dialogue, vision

Muse Wonder 1.0

the Supervisor, change planning with rollback, reports

Voice Transcribe

WhatsApp voice notes become tickets

SAM 3.1

locates objects in camera frames

Try it in two minutes

- Open cloudrium.com/support-agent and type your mobile number or e-mail; the desk sends a one-time code
- You are not in the directory, so give the invitation code from the submission notes and your name: an account is created in Entra ID
- Then write as an employee: "My VPN shows a certificate error", "I forgot my password", "I need a licence for Visio"
- WhatsApp works the same way, voice notes included; every tool call and decision is visible in the ticket

github.com/gokhunyayla/muse-servicedesk · youtu.be/rMqozO4E0ng · Gökhun Yayla